

PYTHON

BASIC → ADVANCED

Data Science Interview Preparation Guide

YOUR LEARNING PATH

Basics → Intermediate → Advanced Python → NumPy → Pandas → Data Cleaning → Visualization → Machine Learning

Interview formula: What is it? → Why use it? → Small example → Common mistake / trade-off

Goal: Build from beginner-level Python to the Python skills commonly used in Data Science interviews.

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Tip: Do not try to memorize every answer word-for-word. Understand the concept and explain it naturally.

■ BASIC PYTHON

Level focus: 31 interview questions. Try answering each question yourself before reading the answer.

Q1. What is Python?

Answer: Python is a high-level, interpreted, general-purpose programming language known for simple and readable syntax.

Q2. Why is Python popular?

Answer: It is easy to learn, has a huge ecosystem, and is widely used in web development, automation, Data Science and AI.

Q3. What is a variable?

Answer: A variable is a name that refers to an object/value, for example: age = 21.

Q4. What are common data types?

Answer: int, float, complex, bool, str, list, tuple, set, dict and NoneType.

Q5. What is dynamic typing?

Answer: Python determines an object's type at runtime, so you normally do not declare the variable type.

Q6. What is type casting?

Answer: Converting a value from one type to another, such as int('10') or str(100).

Q7. What is the difference between == and is?

Answer: == compares values; is checks object identity.

Q8. What is a string?

Answer: An immutable sequence of characters.

Q9. What is a list?

Answer: A mutable, ordered collection that can contain duplicate values.

Q10. What is a tuple?

Answer: An ordered, immutable collection.

Q11. What is a set?

Answer: An unordered collection of unique elements.

Q12. What is a dictionary?

Answer: A collection of key-value pairs.

Q13. What is indexing?

Answer: Accessing an element by its position, such as name[0].

Q14. What is slicing?

Answer: Extracting part of a sequence using start:stop:step.

Q15. List vs tuple?

Answer: Lists are mutable and use []; tuples are immutable and use ().

Q16. append() vs extend()?

Answer: append() adds one object; extend() adds elements from another iterable.

Q17. remove() vs pop()?

Answer: remove() deletes a matching value; pop() removes and returns an element.

Q18. What is if-else?

Answer: It executes different code depending on whether conditions are true or false.

Q19. for vs while?

Answer: for commonly iterates over an iterable; while repeats while a condition is true.

Q20. break vs continue?

Answer: break exits the loop; continue skips the current iteration.

Q21. What is pass?

Answer: A placeholder statement that performs no action.

Q22. What is a function?

Answer: A reusable block of code that performs a specific task.

Q23. print() vs return?

Answer: print() displays a value; return sends a value back to the caller.

Q24. What are parameters and arguments?

Answer: Parameters appear in the function definition; arguments are the actual values passed.

Q25. What is a default argument?

Answer: A parameter with a predefined value.

Q26. What is lambda?

Answer: A small anonymous function, for example `lambda x: x*2`.

Q27. What is list comprehension?

Answer: A compact way to create a list, such as `[x*x for x in range(5)]`.

Q28. What is exception handling?

Answer: Handling runtime errors using try, except, else and finally.

Q29. Why use with open()?

Answer: It safely manages a file and automatically closes it.

Q30. What is a module?

Answer: A Python file containing reusable code.

Q31. What is pip?

Answer: Python's package installer used to install external libraries.

■ **Interview tip:** If the interviewer asks for more detail, give a practical Data Science example.

■ INTERMEDIATE PYTHON

Level focus: 27 interview questions. Try answering each question yourself before reading the answer.

Q32. What are `*args`?

Answer: It collects a variable number of positional arguments into a tuple.

Q33. What are `**kwargs`?

Answer: It collects a variable number of keyword arguments into a dictionary.

Q34. What is `map()`?

Answer: It applies a function to each item of an iterable.

Q35. What is `filter()`?

Answer: It keeps items for which a condition returns True.

Q36. What is `enumerate()`?

Answer: It provides index-value pairs while iterating.

Q37. What is `zip()`?

Answer: It combines multiple iterables position by position.

Q38. What is a generator expression?

Answer: A compact lazy expression that produces values one at a time.

Q39. What is recursion?

Answer: A function calling itself with a base condition.

Q40. What is a class?

Answer: A blueprint for creating objects.

Q41. What is an object?

Answer: An instance of a class.

Q42. What is `__init__()`?

Answer: The initializer method called when an object is created.

Q43. What is `self`?

Answer: A reference to the current object instance.

Q44. What is inheritance?

Answer: A child class reuses or extends a parent class.

Q45. What is encapsulation?

Answer: Bundling data and methods together and controlling access to internal data.

Q46. What is polymorphism?

Answer: The same interface or method can behave differently for different objects.

Q47. What is method overriding?

Answer: A child class provides its own version of a parent method.

Q48. What is abstraction?

Answer: Showing essential behavior while hiding implementation details.

Q49. What is a class variable?

Answer: A variable shared by all instances of a class.

Q50. What is an instance variable?

Answer: A variable belonging to a particular object.

Q51. What is `__str__()`?

Answer: A special method defining a readable string representation of an object.

Q52. What is a custom exception?

Answer: A user-defined exception class for application-specific errors.

Q53. What is a docstring?

Answer: A string used to document a module, class or function.

Q54. What is a higher-order function?

Answer: A function that accepts another function or returns a function.

Q55. What is a shallow copy?

Answer: A copy where nested objects may still be shared.

Q56. What is a deep copy?

Answer: A copy that recursively duplicates nested objects.

Q57. Mutable vs immutable?

Answer: Mutable objects can change after creation; immutable objects cannot.

Q58. `sort()` vs `sorted()`?

Answer: `sort()` modifies a list in place; `sorted()` returns a new sorted result.

■ **Interview tip:** If the interviewer asks for more detail, give a practical Data Science example.

■ ADVANCED PYTHON

Level focus: 20 interview questions. Try answering each question yourself before reading the answer.

Q59. What is an iterator?

Answer: An object that produces values one at a time using `__iter__()` and `__next__()`.

Q60. What is a generator?

Answer: A lazy iterator commonly created with `yield`, producing values one at a time.

Q61. yield vs return?

Answer: `return` ends a function; `yield` pauses a generator and resumes it later.

Q62. What is a decorator?

Answer: A function that wraps another function to add or modify behavior.

Q63. What is a context manager?

Answer: An object used with `with` to manage setup and cleanup of resources.

Q64. What is garbage collection?

Answer: Python automatically reclaims memory from objects that are no longer reachable.

Q65. What is the GIL?

Answer: In CPython, the Global Interpreter Lock allows only one thread at a time to execute Python bytecode.

Q66. Threading vs multiprocessing?

Answer: Threading is useful for many I/O-bound tasks; multiprocessing can use multiple CPU processes for CPU-bound work.

Q67. What is time complexity?

Answer: It describes how an algorithm's resource use grows with input size, often using Big-O notation.

Q68. Why use vectorization?

Answer: It reduces Python-level loops and uses optimized low-level operations for better performance.

Q69. What is a closure?

Answer: A nested function that remembers variables from its enclosing scope.

Q70. What is a property?

Answer: A controlled attribute interface commonly created with `@property`.

Q71. What is a static method?

Answer: A method that does not require `self` or `cls` and is declared with `@staticmethod`.

Q72. What is a class method?

Answer: A method that receives the class as `cls` and is declared with `@classmethod`.

Q73. What is duck typing?

Answer: Python focuses on whether an object supports the required behavior rather than its exact type.

Q74. What is a namespace?

Answer: A mapping from names to objects; Python has local, global, built-in and enclosing namespaces.

Q75. What is LEGB?

Answer: The name lookup order: Local, Enclosing, Global, Built-in.

Q76. What is a virtual environment?

Answer: An isolated environment for project-specific Python packages and versions.

Q77. Why use logging?

Answer: To record program events, warnings and errors for debugging and monitoring.

Q78. What is unit testing?

Answer: Testing small units of code independently to verify expected behavior.

■ **Interview tip:** If the interviewer asks for more detail, give a practical Data Science example.

■ PYTHON FOR DATA SCIENCE

Level focus: 39 interview questions. Try answering each question yourself before reading the answer.

Q79. What is NumPy?

Answer: A library for fast numerical computing with multidimensional arrays and vectorized operations.

Q80. Why is NumPy faster than Python lists?

Answer: NumPy arrays use efficient homogeneous storage and many operations run in optimized compiled code.

Q81. What is ndarray?

Answer: NumPy's multidimensional array object.

Q82. What are shape, ndim and size?

Answer: shape gives dimensions, ndim gives number of dimensions, and size gives total number of elements.

Q83. What is broadcasting?

Answer: NumPy's mechanism for operating on arrays with compatible shapes.

Q84. What is vectorization in NumPy?

Answer: Performing operations on complete arrays without explicit Python loops.

Q85. What is Pandas?

Answer: A library for data manipulation and analysis using Series and DataFrame structures.

Q86. What is a Series?

Answer: A one-dimensional labeled data structure.

Q87. What is a DataFrame?

Answer: A two-dimensional labeled table with rows and columns.

Q88. How do you read a CSV?

Answer: `pd.read_csv('file.csv')`.

Q89. What does info() show?

Answer: Column names, non-null counts, data types and memory information.

Q90. What does describe() do?

Answer: It provides descriptive statistics such as count, mean, standard deviation and quartiles.

Q91. loc vs iloc?

Answer: loc selects by labels; iloc selects by integer positions.

Q92. How do you filter rows?

Answer: Use boolean conditions, such as `df[df['age'] > 25]`.

Q93. How do you handle missing values?

Answer: Detect with `isna()`; handle with `dropna()`, `fillna()`, interpolation or an appropriate model-based approach.

Q94. What is groupby()?

Answer: It groups records so calculations can be performed separately for each group.

Q95. merge() vs concat()?

Answer: merge combines data using matching keys; concat stacks objects along rows or columns.

Q96. What is a pivot table?

Answer: A summarized table that aggregates values across dimensions.

Q97. What is data cleaning?

Answer: Correcting missing, duplicate, inconsistent, invalid and wrongly formatted data.

Q98. What is an outlier?

Answer: An observation unusually far from the typical distribution.

Q99. What is feature engineering?

Answer: Creating or transforming variables to improve analysis or model performance.

Q100. What is normalization?

Answer: Scaling values to a bounded range, commonly 0 to 1.

Q101. What is standardization?

Answer: Transforming a feature using its mean and standard deviation, commonly to mean 0 and standard deviation 1.

Q102. What is data leakage?

Answer: When information unavailable at prediction time enters training or preprocessing.

Q103. What is Matplotlib?

Answer: A foundational Python visualization library.

Q104. What is Seaborn?

Answer: A statistical visualization library built on Matplotlib.

Q105. What is a histogram?

Answer: A chart showing the distribution of a numerical variable.

Q106. What is a scatter plot?

Answer: A chart showing the relationship between two numerical variables.

Q107. What is a box plot?

Answer: A chart showing median, quartiles, spread and potential outliers.

Q108. What is Scikit-learn?

Answer: A machine-learning library for preprocessing, models, evaluation, feature selection and pipelines.

Q109. Classification vs regression?

Answer: Classification predicts categories; regression predicts continuous numeric values.

Q110. What is a train-test split?

Answer: Dividing data into training and testing portions to estimate generalization.

Q111. What is overfitting?

Answer: When a model performs very well on training data but poorly on unseen data.

Q112. What is underfitting?

Answer: When a model is too simple to capture important patterns.

Q113. What is a Pipeline?

Answer: A sequence of preprocessing and modeling steps treated as one workflow.

Q114. What is cross-validation?

Answer: Evaluating a model over multiple train/validation splits.

Q115. Accuracy vs precision vs recall?

Answer: Accuracy is overall correctness; precision focuses on predicted positives; recall focuses on actual positives found.

Q116. What is F1-score?

Answer: The harmonic mean of precision and recall.

Q117. Why use random_state?

Answer: To make randomized experiments reproducible.

■ **Interview tip:** If the interviewer asks for more detail, give a practical Data Science example.

■ FINAL RAPID REVISION

Topic	Remember
List	Mutable, ordered, duplicates allowed
Tuple	Immutable, ordered
Set	Unique elements
Dictionary	Key-value pairs
Function	Reusable block of code
*args	Variable positional arguments
**kwargs	Variable keyword arguments
OOP	Class, object, inheritance, encapsulation, polymorphism, abstraction
Generator	Lazy values using yield
Decorator	Wraps/modifies function behavior
NumPy	Fast numerical arrays
Pandas	Data manipulation and analysis
Matplotlib/Seaborn	Data visualization
Feature engineering	Create useful model features
Pipeline	Connect preprocessing + model
Overfitting	Good training, poor unseen performance
Data leakage	Future/unavailable information enters training
Precision	Of predicted positives, how many are correct
Recall	Of actual positives, how many are found

Final interview strategy: Start with a simple definition. Then explain why it is useful. If possible, give one practical example and mention an important limitation. This makes your answer sound clear and professional.